Change Captioning: A New Paradigm for Multitemporal Remote Sensing Image Analysis

Genc Hoxha, *Graduate Student Member, IEEE*, Seloua Chouaf, *Member, IEEE*, Farid Melgani, *Fellow, IEEE*, and Youcef Smara

***Abstract*— Change detection (CD) is among the most important applications in remote sensing (RS) that allows identifying the changes that occurred in a given geographical area across different times. Even though CD systems have seen a lot of progress in RS, their output is either a binary map highlighting the changing area or a semantic change map that indicates the type of change for each pixel. The change maps are often difficult to interpret by end users, and they omit important information such as relationships and attributes of the changed areas. Motivated by the recent advancement of image captioning in the RS community, in this article, we propose to describe the changes over bitemporal images through change sentence descriptions. The aim of this article is to provide a user-friendly interpretation of the occurred changes. To this end, we propose two change captioning (CC) systems that take bitemporal images as input and generate coherent sentence descriptions of the occurred changes. Convolutional neural networks (CNNs) are used to extract discriminative features from the bitemporal images and recurrent neural networks (RNNs) or support vector machines (SVMs) are exploited to generate coherent change descriptions. Furthermore, in the absence of a CC dataset to test our systems, we propose two new datasets. One is based on very high-resolution RGB images, and the other one is based on multispectral RS images. The obtained experimental results show promising capabilities of the proposed systems to generate coherent change descriptions from the bitemporal images. The datasets are available at the following link: https://disi.unitn.it/∼melgani/datasets.html.**

***Index Terms*— Change captioning (CC), change detection (CD), convolutional neural networks (CNNs), image captioning (IC), recurrent neural networks (RNNs), support vector machines (SVMs).**

I. Introduction

OBSERVING Earth surface evolution is one of the main purposes of remote sensing (RS). Through multitemporal images acquired by sensors onboard satellites or aerial platforms, we can continually observe and track changes that occur around our globe. As a consequence, change detection (CD) is among the most important applications in RS. The task of CD is to automatically detect changes over time in a given geographical area by analyzing two or more co-registered images [1], [2]. Singh [1] defined CD as "the process of identifying differences in the state of an object or phenomenon by observing it at different times". Accordingly, the CD is indispensable in a wide range of RS applications based on land cover and land use analysis such as urban planning, environmental change assessment, disasters monitoring, deforestation, and agricultural investigations [1].

In the RS literature, several approaches have been proposed for CD. These approaches can be grouped into two main broad categories: 1) unsupervised and 2) supervised approaches. Unsupervised approaches include CD methods that do not rely on prior information provided by reference data. In contrast, supervised approaches need the *a priori* information of reference data as most of them are based on the use of supervised classifiers. The majority of the unsupervised approaches in the literature are based on the so-called "difference image" (DI) generation and analysis [3]–[10]. DI can be generated by applying pixel-by-pixel subtraction [3] or ratio operator [5] in the co-registered input images where the assumption is that the pixels associated with the land cover changes have different values compared to the ones associated with the unchanged areas. Threshold or clustering methods can be used to analyze the DI and produce the final change map [2]–[4]. In general, unsupervised CD approaches are appealing because they do not need ground-truth information. However, defining the optimal threshold to produce the final change map is not an easy task, and its effectiveness strongly depends on the statistical characteristics of the DI [4], [6]. Furthermore, the outcome change map is, in most of the cases, binary (change/no change) [2]–[10]. Even though there are specific cases in which different kinds of changes can be detected, the exact land-cover transition associated with the change cannot be explicitly identified since the ground-truth information is missing [2].

Supervised approaches assume that the ground-truth information is available for all the multitemporal images. This information is injected into machine-learning algorithms to learn automated decision models to distinguish between changed and nonchanged areas and possibly the type of change in the multitemporal images [11]–[20]. This is achieved thanks to the use of supervised classifiers such as support vector machines (SVMs) [18], [21], [22], decision trees, and random forest [23]. Postclassification comparison is a traditional supervised technique that independently classifies the multitemporal images and then compares the individual classification maps to define the changed areas and the exact land-cover

Manuscript received 7 February 2022; revised 28 April 2022 and 6 July 2022; accepted 24 July 2022. Date of publication 1 August 2022; date of current version 12 August 2022. *(Corresponding author: Farid Melgani.)*

Genc Hoxha and Farid Melgani are with the Department of Information Engineering and Computer Science, University of Trento, 38123 Trento, Italy (e-mail: genc.hoxha@unitn.it; melgani@disi.unitn.it).

Seloua Chouaf and Youcef Smara are with the LTIR Laboratory, University of Sciences and Technology Houari Boumediene, Algiers 16111, Algeria (e-mail: chouaf.seloua@gmail.com; yousmara@yahoo.com).

Digital Object Identifier 10.1109/TGRS.2022.3195692

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:31:04 UTC from IEEE Xplore. Restrictions apply.

transitions associated with the change [12], [13], [15], [18]. The advantage of this technique is that it is straightforward. However, postclassification comparison techniques suffer the propagation of the classification error which leads to an error accumulation in the final change map [1]. To alleviate the classification error accumulation present in postclassification comparison techniques, compound classification simultaneously analyzes the multitemporal data by taking into account temporal contextual information. Markov and conditional random fields have shown to be very effective in incorporating such information and improving the CD results in terms of accuracy and reliability [11], [14], [16], [19]. Although supervised CD approaches are very effective in analyzing the multitemporal images and detecting the changes present, they assume that the ground truth information is available for each of the multitemporal images. Such an assumption is not always possible in real scenarios as collecting ground-truth information is time-consuming and costly. To address this, semisupervised CD approaches assume that the labeled information is available for at least one of the multitemporal images and the learning paradigm of these approaches benefits both labeled and unlabeled samples making them more suitable in real-life scenarios [24], [25].

One common limitation of the aforementioned conventional CD approaches is the use of hand-crafted features. These features are tedious to obtain and unable to capture high-level feature representation for multitemporal images. Recently, deep neural networks have shown to be very effective in automatically learning discriminative and representative features directly from raw images. Improvement in the quality and quantity of RS data along with more accessible computational resources has oriented the researchers in the RS community toward the exploitation of deep learning (DL) algorithms to improve the accuracy of CD systems [26]–[29]. The DL algorithms exploited for CD include restricted Boltzman machines [30], autoencoders [31], convolutional neural networks (CNNs) [32]–[39], and recurrent neural networks (RNNs) [40]. A comprehensive and extensive review of these strategies can be found in [29].

Despite the efforts made in developing reliable and accurate CD systems, the obtained output is either a binary change map that highlights the changed regions, or a semantic change map that indicates the type of change for each pixel. The improvement of RS technology has enabled the acquisition of high-spatial resolution images characterized by abundant spatial information. Hence, a pixel-level interpretation of the changes (change map) not only can be unsuitable but also onerous. Furthermore, important and abundant high-level semantic information presents in high-spatial resolution such as the attributes and the relationships of the changed areas are omitted by the change maps. This information needs to be inferred and interpreted directly by end users. It is worth noting that the interpretation of change maps might be challenging for end users who do not have expertise in the topic. User-friendly information about the changed areas would widen the impact of CD applications in the community. To address this issue, in this article, we present a CD system that automatically describes the changes in bitemporal images with sentence descriptions (i.e., captions). In doing so, we are inspired by the recent advancements of image captioning (IC) in the RS community [41]–[53]. IC is a cross-disciplinary technique between computer vision and natural language processing (NLP) that aims at analyzing and describing the visual content of an image with natural sentence descriptions [54], [55]. Compared to traditional RS image analysis techniques, like classification or object recognition, IC not only provides a more user-friendly representation but also a richer one including relationships and attributes of the detected objects. High-level semantic information such as relationships and attributes of objects present in an RS scene has shown to be very important in a variety of applications such as image retrieval [56] and generation [57]. In this work, we aim at expanding the remote sensing image captioning (RSIC) systems from single-date images to bitemporal images where the goal is transformed from describing the image content to describing the changes that have occurred between the two acquisitions.

From a methodological point of view, RSIC systems can be divided into three groups [41]–[53]: 1) template-based; 2) retrieval-based; and 3) encoder–decoder frameworks. Template-based and retrieval-based systems are the most naïve ones. The former assumes the availability of fixed sentence templates with empty slots, and object detection algorithms are used to fill the slots with detected objects to form a complete sentence [42]. The latter assumes an archive where a candidate reference description for a target image is always found. The captioning system is treated as a retrieval problem, where for a target image, the best candidate caption is retrieved from the archive [45]. Template-based methods are tedious and tend to be simple and nonnatural compared to human ones due to the fixed templates. Retrieval-based methods do not generate novel captions, and as a consequence, the content of the retrieved caption might be irrelevant to the target image.

The most used captioning strategies in the RS community are the encoder–decoder based methods and their attention variants. These methods are inspired by the progress of DL and machine translation [58], [59]. In general, CNNs are used as encoders to extract discriminative features and sequential models, like RNNs, are used as decoders to translate the features into a sentence description. The advantage of these methods is their ability to generate coherent and human-like descriptions. Qu *et al.* [41] proved for the first time the effectiveness of the encoder–decoder framework in generating descriptions for RS images. Lu *et al.* [43] explored different RSIC models based on hand-crafted features [60]–[62] along with deep features [63] and attention mechanisms. They also released the biggest dataset for RSIC that gave the opportunity to different researchers to develop and test many RSIC algorithms [44], [46], [48]–[53]. For instance, an RSIC system based on attribute attention mechanism that fuses local and global information is proposed in [44]. Wang *et al.* [45] proposed a denoising-based multiscale feature fusion to deal with the large-scale variation in the RS images. An active attention mechanism and a retrieval topic recurrent memory where the sound of different objects and sentence topic are used as guiding information to generate the descriptions, respectively, are proposed in [49] and [50]. Sumbul *et al.* [51] proposed a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:31:04 UTC from IEEE Xplore. Restrictions apply.

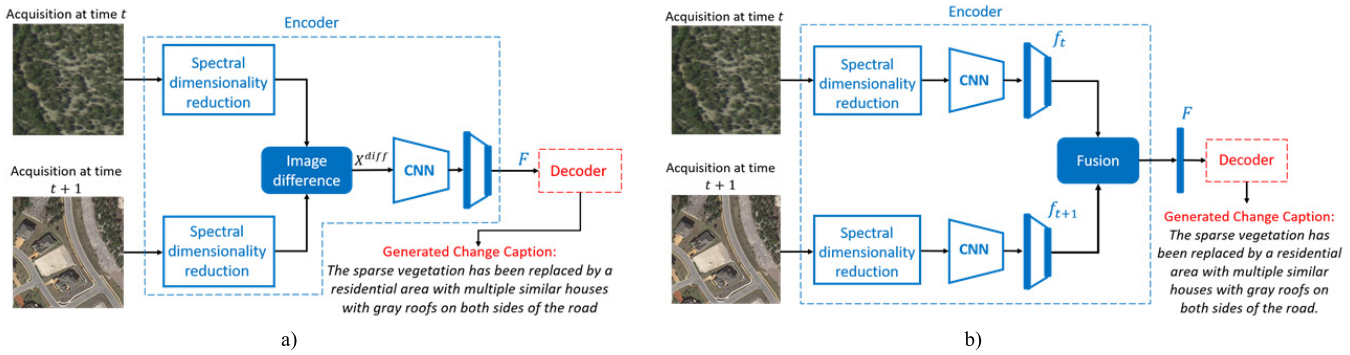


Fig. 1. Overview of the proposed CC system based on: (a) image-based level fusion and (b) feature-based level fusion.

summarization-driven RSIC system where a pretrained pointer generator [64] summarizes different ground-truth captions and injects this information into the captioning system through an attention mechanism. To alleviate the problem of overfitting, a truncation loss entropy is presented in [52] and a decoder based on SVMs is presented in [53].

The aforementioned RSIC systems deal with caption generation for single-date RS images where the goal is to describe the content of the considered image. In this work, we propose a change-captioning (CC) system that aims at describing the changes that occurred between two different acquisitions of the same geographical area. To this end, the system takes bitemporal images as input instead of single-date images and describes the possible changes. As shown in Fig. 1, it is based on an encoder–decoder framework. The encoder is composed of three main blocks: 1) a spectral dimensionality reduction block that aims at reducing the spectral dimension of the images; 2) a fusion block that combines the multitemporal information; and 3) a pretrained CNN to extract discriminative features from the multitemporal images. We propose two main fusion strategies: early fusion which is applied at image level (before the CNN) and a late fusion which is applied at feature level (after the CNN) as it can be seen in Fig. 1(a) and (b), respectively. Once the encoding process is done, the obtained bitemporal image representations are forwarded to the decoder to generate coherent change descriptions. In particular, we implement two different decoders: the first one is based on the classical RNNs. The second one relies on a network of k multiclass SVMs, where the last one is rendered recurrent. Fig. 2 shows the block schemes of the two proposed decoders. To test the proposed remote sensing change captioning (RSCC) systems, in this work, we have built two different datasets. The first one is based on very high resolution (VHR) images while the second one is based on medium resolution satellite images. Both datasets contain 500 images and, to each image, are assigned five different descriptions. The two datasets will be made publicly available to boost research in this new area.

The main contribution of our work can be summarized as follows.

1) We propose two baseline methods for RSCC based on the encoder–decoder framework. The difference of the baseline methods consists in the decoder part. A classical RNN is developed in the first baseline method while a decoder based on the SVMs is proposed in the second baseline method.
2) Two datasets for RSCC are created. The first one consists of VHR images while the second one consists of medium-resolution satellite images. Each dataset is composed of 500 bitemporal images and each bitemporal image is assigned five different change descriptions.

We have briefly presented an RSCC system in [65]. This article extends our previous work [65] by proposing novel RSCC methods and disclosing the two datasets.

The rest of this article is organized as follows. Section II introduces the two proposed CC datasets (CCDs). Section III describes the proposed RSCC systems. Section IV shows the experimental results, and finally, Section V concludes this work.

II. Proposed CCDs

In this section, we propose two novel datasets to caption the changes of the bitemporal RS images. It is worth noting that these datasets are the first ever made to tackle the problem of CD through textual descriptions that carefully summarize the changes of a given geographical area between two different acquisitions. The datasets are different between each other. The first dataset is obtained by exploiting and annotating an existing dataset for RS CD. The considered dataset is LEVIR and is originally used for detecting changes related to buildings [66]. The second dataset is entirely built in this work and consists of multispectral Landsat 7 images acquired over the region of Dubai. In the following, we describe each of the dataset and the annotation process.

A. LEVIR CCD

The LEVIR dataset is originally constructed for building CD and consists of 637 VHR (0.5 m/pixel) bitemporal images of dimension 1024 × 1024 acquired over 20 different regions over different cities in TX, USA [66]. The collection of the image pair is obtained from Google Earth API with a time span of 5–14 years. A full description of the original dataset for building CD and the terms of use are given by Chen and Shi [66].

To make the dataset appropriate for RSCC, we have exploited and annotated it with sentence descriptions that

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:31:04 UTC from IEEE Xplore. Restrictions apply.

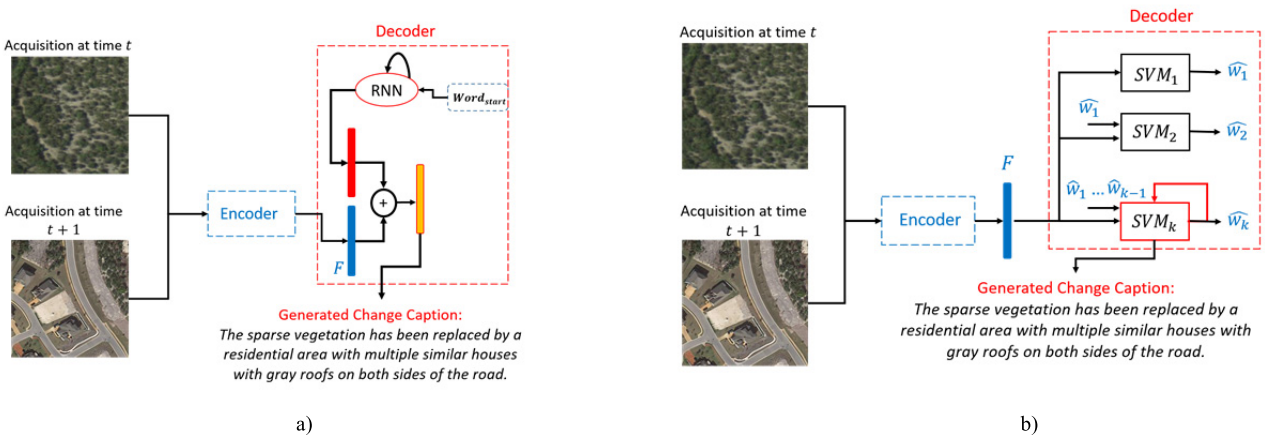


Fig. 2. Overview of the decoding phase based on: (a) RNNs and (b) SVMs.

describe the nature of changes occurred between the two acquisitions. In particular, the annotation process is performed by annotators with RS background. First, a visual inspection of the scenes in the bitemporal images is performed to identify the nature of changes present. Then, the attributes (i.e., color and size) and relationships between different changes are considered, in terms of position and relative position to other objects. After, the original image pairs are cropped into 256 × 256 pixels patches and finally, to each image pair, five different change sentence descriptions are given by the annotators. The total number of image pairs is 500 leading to a total number of 2500 change descriptions.

From the visual inspection, we have identified 13 types of objects that undergo a change. These objects include: sparse/dense vegetation area, bare land, residential area, houses, roofs, roads/streets, vehicles, constructions, trees, grass, and swimming pools. The relationships between the objects and their positions and relative positions mainly include the directions such as: on the right/left, on the top or bottom part, and on the top left/right and bottom left/right. The attributes of the objects include: color (i.e., red, gray, and white), the size (big/small), and their similarity (similar/different). Starting from the visual inspections, the annotators are asked to formulate a coherent change sentence description including the attributes and the relationships of different objects. In the dataset, the minimum length of the sentence is 3 and it refers to a situation in which there is no change between the two acquisitions: "nothing has changed." The maximum length of the sentence is 37 words while the average is 15.12 words. The total vocabulary size of the dataset is 279 unique words. Two examples from the dataset are depicted in Fig. 3. The terms of use for the dataset follow [66].

B. Dubai CCD

From 2000 to 2010, the area of Dubai has significantly changed in terms of urbanization. To understand and automatically describe the urbanization phenomena in Dubai, we decided to create the Dubai CCD. To this end, we needed images of the year 2000 and 2010. From our research, the only free available images that cover the area of Dubai in the time of interest are multispectral images acquired by Enhanced Thematic Mapper Plus (ETM+) sensor onboard Landsat 7 [67]. ETM+ acquires images in different bands with different spatial resolution: 30-m visible and near infrared bands; 60-m thermal band; and 15-m panchromatic band [67]. Fig. 4 shows the RGB representation of the images. The bitemporal images are acquired on May 19, 2000 and on June 16, 2010, respectively, and have dimension of 2569 × 2468 pixels. To create the Dubai CDD, we have considered only visible and infrared bands which have a spatial resolution of 30 m. Considering the medium spatial resolution of the images here, we extracted 500 tiles of sizes 50 × 50 from the bitemporal images. In order to properly identify and describe the nature of the changes in the small bitemporal tiles, we also resorted to Google Maps to better inspect the area and publically available documents to understand the physical composition of the area. From the visual inspection, we identified six different broad categories: 1) roads (i.e., crossroads and roundabout); 2) houses (i.e., neighborhood and residence areas); 3) buildings (any structure different from house); 4) green areas; 5) lakes; and 6) islands (i.e., archipelagoes and peninsula). The annotators were asked to describe the changes on the bitemporal images with a minimum of three words accounting for spatial distribution and the attributes of the changes. Each image was annotated by five different change descriptions yielding to 2500 descriptions, a maximum length of 23 words, an average length of 7.35 words, and a vocabulary size of 246 unique words. Fig. 5 depicts two examples from the dataset along with the reference descriptions.

III. Proposed CC Systems

Let $X_i = (I_t, I_{t+1})$ be a pair of RS images (or patches) acquired over the same geographical area at time t and $t+1$. Let us assume that we have M pairs of such images $X = \{X_i\}_{i=1}^{M}$ and each pair is associated at least one textual description (sentence/caption) that describes the most salient changes that have occurred between the two acquisitions. Let $S_i = [w_1, w_2, \ldots, w_L]$ be the sentence change description of the image pair X_i composed of L ordered words

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:31:04 UTC from IEEE Xplore. Restrictions apply.



1. Many similar houses with gray roofs appear on the right instead of the bare land.
2. The bare land on the right has been replaced by many similar houses with gray roofs.
3. Many trees appear near the houses present on the left and no vehicle remains on the right road.
4. New roads and houses have been created, many trees appear near the houses and different vehicles are present.
5. Multiple similar house with gray roofs has been added on the sides of the existing road on the right of the residential area.

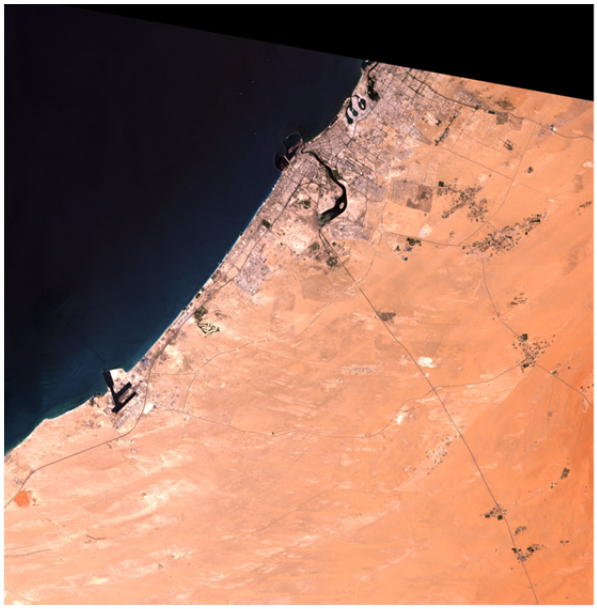
a)



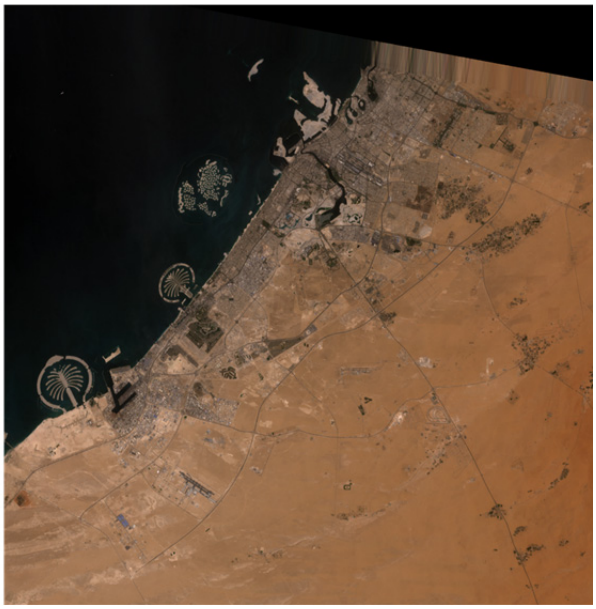
1. The grass appears more green and the trees have increased in size.
2. There is no change except the color of the grass and the size of the trees.
3. Only the trees and the grass appear different.
4. Slight changes related to the color of the grass and the size of the trees.
5. There is no significant change to mention expect for the grass that have grown.

b)

Fig. 3. Two examples from LEVIR CCD along with the reference descriptions: (a) significant change and (b) slight changes between the acquisitions.



a)



b)

Fig. 4. Images acquired over the region of Dubai from Landsat 7 on: (a) May 19, 2000 and (b) June 16, 2010. For visualization purposes, the RGB combination is shown.

w. Inspired by the captioning systems developed for single images, our CC systems are composed of two main steps: 1) deep bitemporal image feature extraction and fusion (i.e., encoder) and 2) sentence generation (i.e., decoder). The first step aims to fuse and represent the bitemporal images with discriminative features while the second one targets the translation of the features into a coherent textual description that describes the nature of the changes detected among the two acquisitions. The encoder is composed of two branches in the case of bitemporal images (and more branches in the case of multitemporal images). Each branch conveys three main blocks: 1) spectral dimension reduction to reduce the spectral dimension of the images; 2) a CNN to extract discriminative features; and 3) a fusion block to integrate the bitemporal (or multitemporal) information. The fusion operation can be applied either at image level before the CNN or at feature level after the CNN. To generate the sentences that describe the possible changes, we exploit both DL-based methods such as RNNs, as well as the recently introduced SVM-based decoder [53]. Figs. 1 and 2 depict the proposed captioning system according to different configurations of the encoder and decoder.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:31:04 UTC from IEEE Xplore. Restrictions apply.



Fig. 5. Two examples from Dubai CCD along with the reference descriptions: (a) significant change and (b) slight changes between the acquisitions.

A. Multitemporal Image Representation and Fusion

1) Spectral Dimensionality Reduction: This is the first block of the encoder. It is used to reduce the spectral dimensionality of multispectral images. In particular, we rely on the simple and popular principal component analysis (PCA) transformation [67] to reduce the original spectral dimension into three channels to fit the pretrained CNN input requirement.

2) Multitemporal Image Representation: For the scope of feature generation, we rely on the powerful VGG16 CNN architecture pretrained on ImageNet [68] to extract discriminative features [63]. The image features are obtained passing the input information from the bitemporal images through the pretrained VGG-16 (omitting the last fully connected layer) as follows:

$$f_i = \text{VGG16}(X_i). \tag{1}$$

3) Multitemporal Image Fusion: Different from single IC, our scenario is composed of bitemporal images. To encode the changes, we consider two different fusion strategies as shown in Fig. 1. In the first one, the fusion is done before the feature extraction process (early fusion). We apply channelwise image difference $X_i^{\text{diff}} = (I_{t+1} - I_t)$, and afterward, we forward the DI to the CNN to extract the image features. In the second one, fusion is performed once the individual features are extracted by the pretrained CNN (late fusion). In this configuration, we have considered two simple but very effective operators, feature concatenation $F = (f_t, f_{t+1})$ or elementwise feature subtraction $F = (f_{t+1} - f_{t-1})$.

B. Sentence Generation

A sentence is a set of ordered words correlated with each other within a semantic context. To form a sentence, we need the knowledge of the previous words. We can imagine the sentence generation as a sequential process in which the words are selected on the basis of the previous ones. Hence, the automatic generation process of a sentence could be formulated in a sequential way in which the words would be generated once at a time on the basis of the previously generated ones.

In the NLP community, RNNs have been shown to be a default choice when dealing with sequential data. RNNs are composed of feedback loops that act as memory to store and use past information to generate coherent and complete sentences. Being RNNs part of the broad DL family, they need a large training dataset to have good performances. In the RS community, obtaining a large amount of training samples is not always possible. The annotation process is very expensive in terms of time and resources. To cope with this problem, a novel decoder based on SVMs has recently been introduced in the RS community for IC [53]. SVMs are well-known classifiers that are based on margin maximization principles which render them more robust to overfitting compared to DL methods [21], [22]. The SVM decoder is composed of k multiclass classifiers in a cascade where the last one is rendered recurrent to model the dependence on the previous information while describing the changes between different acquisitions [53].

In the following, we describe the decoder based on RNNs and SVMs and their exploitation with the bitemporal images to generate the captions of the changes of different acquisitions.

1) CC System With RNNs as Decoder: Decoders in the form of RNN are very popular for solving IC problems. Inspired by the findings of [69], in this work, we use the RNN to deal with the sentence encoding only, and subsequently, through a multimodal layer, we give the image information to bias the sentence generation process to describe the changes of the bitemporal images. We noted that for relatively small datasets, this configuration leads to better performances.

Let x_t be the one-hot encoding that represents a given word $w_t \in V$ where V is the vocabulary (whose size is the total number of words). Let h_t be the RNNs' memory. At each time step, RNNs take as input x_t and produces an output y_t which is a combination of h_{t-1} and the current input x_t. The following equations describe the RNN, where e_t is the so-called embedding layer which projects the one-hot

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:31:04 UTC from IEEE Xplore. Restrictions apply.

encodings of the words into a semantic space where words having similar meaning are scattered near to each other, g_1 is the elementwise logistic sigmoid function, U, V, W represent the weights to be learned, and b is the relative bias:

$$e_t = W_x x_t \tag{2}$$
$$h_t = g_1(U_w e_t + U_h h_{t-1} + b) \tag{3}$$
$$y_t = V_h h_t. \tag{4}$$

The output is then combined through a multimodal layer with the bitemporal features obtained by VGG16 as follows:

$$m_t = g_2\left(U_y y_t + U_F F_i + b\right) \tag{5}$$

where g_2 is the rectified linear unit (ReLU). The last layer of the CC model consists of a fully connected layer of size V (vocabulary size) with softmax activation function [see (6)] which estimates the probability of the next word of the sentence given the combination of the image features and the previously predicted words

$$s(m_t) = e^{m_t} \Big/ \sum_{k=1}^{V} e^{m_k}. \tag{6}$$

In the training phase, the model receives the bitemporal images $X_i = (I_i, I_{i+1})$ along with one of the reference change descriptions and the goal of the model is to learn the parameters θ that maximize the probability of constructing the correct sentence that describes the changes between different acquisitions as follows:

$$\theta^* = \underset{\theta}{\operatorname{argmax}} \sum_{X_i, S} \log p(S \mid X_i; \theta). \tag{7}$$

Decomposing the sentence descriptions into words, (7) becomes

$$\log p(S \mid X_i) = \sum_{l=1}^{L} \log p(w_k \mid X_i, w_1, \ldots, w_{k-1}) \tag{8}$$

where $p(w_k \mid X_i, w_1, \ldots, w_{k-1})$ is the probability of predicting w_k (kth word) given the image information and the previously predicted words. Two special tokens are inserted to each sentence to account for the start and the end of a sentence. Finally, the best parameters of the model are estimated by minimizing the following standard cross-entropy loss function:

$$\text{Loss}(X, S) = -\sum_{i=1}^{l} \log p_i(w_i). \tag{9}$$

In the test phase, the bitemporal images are given as input to the model to generate the change caption highlighting the relevant changes that have occurred between the two acquisitions. The prediction process ends when the special token indicating the end of the sentence is predicted.

Different studies have shown that the simple RNNs are affected by gradient vanishing/exploding problems, which means that RNNs tend to forget faraway previous information that might be relevant for the prediction of subsequent words. To overcome this problem, researchers have come with long short-term memory (LSTM) [70] and their variation gated recurrent unit (GRU) [71]. These special RNNs are specifically designed to cope with gradient vanishing/exploding problems. They have a more complex internal structure composed of different gates that allow the backpropagation of the gradients through time with uninterrupted gradient flow [70], [71]. In particular, in our CC, we have used GRU to generate descriptions of the changes. The GRU architecture is illustrated in Fig. 6. The information flow in GRU is controlled by two gates, a reset gate r_t that decides how much of previous information stored in h_{t-1} to keep and an update gate z_t that combines the previous information with the new input to update the new state h_t. Equation (3) changes to the following equations that describe the GRU network:

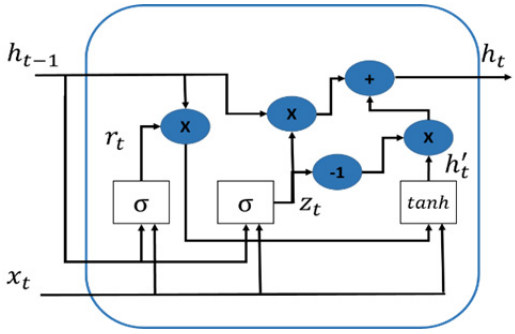


Fig. 6. GRU architecture.

$$h_t = z_t \cdot h_{t-1} + (1 - z_t) \cdot h'_t \tag{10}$$
$$z_t = \sigma\left(W_z\left[e_t, h_{t-1}\right] + b\right) \tag{11}$$
$$h'_t = \tanh\left(W_h\left[e_t, r_t \cdot h_{t-1}\right] + b\right) \tag{12}$$
$$r_i = \sigma\left(W_r\left[e_t, h_{t-1}\right] + b\right). \tag{13}$$

2) CC System With SVM-Based Decoder: The decoder consists of k SVM linear multiclass classifiers in cascade where the last one, namely, SVM_k, is rendered recurrent to model the prediction of a new word given the bitemporal image features and the previously predicted words [53]. In the training phase, each SVM is responsible to learn the mapping between the previously predicted words (if any) of the change sentence description and the related bitemporal image feature on one side, and the subsequent word(s) on the other. The following equations describe this process:

$$w_1 = \text{SVM}_1(F_i) \tag{14}$$
$$w_2 = \text{SVM}_2(F_i, w_1) \tag{15}$$
$$w_{k-1} = \text{SVM}_{k-1}\left(F_i, w_{k-2,}\right). \tag{16}$$

The kth SVM classifiers are recurrent. It learns the mapping between the previous $k-1$ words and image feature on the one hand and the subsequent $L-K$ words, on the other hand. The words are encoded using one-hot encoding. Image features and word features are concatenated together to form the input of each SVM classifier. This corresponds to the SVM-D CONC configuration [53].

At test time, the bitemporal image is given as input to the SVM decoder to generate the CC description. The equations describing the sentence generation at testing time are as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:31:04 UTC from IEEE Xplore. Restrictions apply.

follows:

$$\widehat{w_1} = \text{SVM}_1(F_i). \tag{17}$$

$$\widehat{w_2} = \text{SVM}_2\left(F_i, \widehat{w_{1,t}}\right). \tag{18}$$

$$\vdots$$

$$\widehat{w_k} = \text{SVM}_k\left(F_i, \widehat{w_{k-1}}, \ldots, \widehat{w_1}\right). \tag{19}$$

The recurrent SVM (SVM_k) in recurrent way will complete the sentence generation on the basis of the previously $k-1$ predicted words (and on the image features). The generation process ends when predicting the last word that indicates the end of the sentence.

IV. Experiments

A. Experimental Setup

In this section, we evaluate the two RSCC systems on the proposed LEVIR and Dubai RSCC datasets. The split of the datasets is as follows: 60%, 10%, and 30% for training, validation, and testing, respectively. We obtain the image features using VGG-16 pretrained on ImageNet. VGG-16 produces a fixed feature vector of dimension 4096. In this work, we have considered two different configurations of the systems based on how image processing is performed. In the first configuration, we first obtain a DI by applying channelwise difference of the individual images composing the bitemporal image and then apply the CNN to extract the features. In this configuration, the obtained feature vector is of 4096 dimension. In the second configuration, the individual image features are extracted from each bitemporal image and later feature fusion is applied. The fusion operator is either concatenation, doubling the size of the feature vector, or elementwise subtraction of the individual feature vectors obtaining a fused feature vector of dimension 4096. To the system based on the RNNs, the original features are projected through a projection layer (dense layer) with activation function ReLu and dimension 256. The same dimensions are fixed for the word embedding layer and the internal memory of the RNN. The output of the RNN and the image features are fused together applying elementwise addition. Adam optimizer [72] with a learning rate of 0.0001, cross-entropy loss function, and a batch size of 128 are used to train the model. Dropout regularization is applied in different parts of the model to avoid overfitting. In particular, dropout with a rate of 0.5 is applied to the original image features, to the embedding layer, and to the output of the RNN. The model is trained for 150 epochs. However, early stopping is applied if the validation accuracy does not improve. The experimental results in terms of accuracy, training, and testing times are based on the model that achieves the highest validation accuracy (see Section IV-C). Note that this configuration yields the best results. We also tried to concatenate the image features and the words features forming the multimodal layer without applying any reduction and the results were poor.

The SVM decoder is composed of K multiclass classifiers. In our experiments, we adopted $K = 6$, for LEVIR CCD and $K = 4$ for Dubai CCD, respectively. This choice depends on the average length of the change sentence descriptions which are 15.12 and 7.35 for LEVIR CC and Dubai CCDs, respectively. Because of the intrinsic properties of the SVM decoder, we need the number of SVM to be almost half of the average sentence length to model the word dependency while generating the change captions. This means that, in total, the system is composed of six and four SVM multiclass classifiers for LEVIR CCD and Dubai CCD, respectively. The last one is rendered recurrent to model the dependence of the generated change caption on the previously predicted words. In particular, we rely on linear SVM multiclass classifier that has only one free parameter, the value of the regularization C. The best values found on validation set are $C = 0.001$ for SVM_1 and $C = 0.01$ for the rest of SVMs. Each SVM takes as input the concatenation between the bitemporal image features and words features (if present). In contrast to the CC system based on the RNN, no reduction is applied to the image feature vector. This is well-handled by the SVM that tolerates high-dimensional inputs. Following [53], all image and words features are scaled to be in the range [0, 1] using minimax scaler characterized by the following formula:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{20}$$

where x is the original feature value, $x_{\min}$ and $x_{\max}$ are the minimum and maximum feature values, respectively, and x' is the normalized feature value. The experiments were conducted on an Intel Xeon CPU E5-1620 v3 at 3.50-GHz machine.

B. Evaluation Metrics

Following the RSIC works, the metrics used to evaluate the performances of our RSCC systems are bilingual evaluation understudy (BLEU) [73], recall-oriented understudy for Gisting evaluation (ROUGE_L) [74], metric for evaluation of translation with explicit ordering (METEOR) [75], and consensus-based image description evaluation (CIDEr) [76]. They measure how close the outputs of a CC system (generated change description) are to the descriptions provided by human experts (reference change descriptions). In particular, BLEU score uses the n-gram (n-consecutive words) precision to quantify the similarity between the generated change caption and the reference ones. In this article, n ranges from 1 to 4. ROUGE_L calculates the F-score with respect to the longest common subsequence between the generated descriptions and the reference ones. METEOR is based on the harmonic mean of unigram precision and recall, with recall weighted higher than precision. Finally, CIDEr computes the similarity between a generated description and reference descriptions based on the term frequency inverse document frequency (TF-IDF). The main idea of CIDEr is to weigh in a different way the words present in the corpus. The most frequent words are likely to be characterized by low information while the rarest ones have more information.

C. Experimental Results on Different Datasets

1) Quantitative Results: Tables I and II report the quantitative results of the two proposed methods on LEVIR and Dubai CCDs, respectively. In particular, in Tables I and II, we can see

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:31:04 UTC from IEEE Xplore. Restrictions apply.

TABLE I
EVALUATION SCORES (%) AND TIMES NEEDED FOR TRAINING/TESTING ON LEVIR CCD

Method	Fusion operator	B-1	B-2	B-3	B-4	METEOR	ROUGE-L	CIDEr	Training Time (minutes)	Testing Time (seconds)
CC System based on RNN	Feature Concatenation	73.19	61.70	52.77	45.88	26.07	52.94	59.97	29.9	16.92
	Feature Subtraction	71.85	60.40	52.18	45.94	**27.43**	**54.13**	**71.64**	48.3	19.29
	Difference Image	68.20	54.95	45.64	38.79	25.34	49.86	60.27	48.75	16.89
CC System based on SVM	Feature Concatenation	70.89	60.19	51.59	44.80	24.19	50.60	54.87	9.88	2.36
	Feature Subtraction	**73.83**	**63.18**	**54.10**	**46.97**	25.09	50.48	51.42	36.4	**1.21**
	Difference Image	66.94	54.81	45.25	37.99	22.69	45.61	42.49	**2.58**	1.29

TABLE II
EVALUATION SCORES (%) AND TIMES NEEDED FOR TRAINING/TESTING ON DUBAI CCD

Method	Fusion operator	B-1	B-2	B-3	B-4	METEOR	ROUGE-L	CIDEr	Training Time (minutes)	Testing Time (seconds)
CC System based on RNN (RGB channels)	Feature Concatenation	64.41	49.39	38.24	28.03	25.85	50.65	68.85	12.4	8.47
	Feature Subtraction	67.19	52.23	40.00	28.54	25.51	51.78	69.73	16.57	8.23
	Difference Image	65.54	51.51	40.13	28.70	25.34	51.90	69.57	15.24	11.52
CC System based on RNN (PCA)	Feature Concatenation	**71.44**	56.80	45.36	34.38	**28.07**	**56.67**	80.73	10.25	5.57
	Feature Subtraction	70.71	**57.58**	**46.10**	**35.50**	27.60	56.61	**82.96**	13.69	5.53
	Difference Image	64.84	49.98	39.57	30.22	24.36	51.57	67.69	12	8.16
CC System based on SVM decoder (RGB channels)	Feature Concatenation	65.78	51.25	40.34	30.74	24.94	50.66	69.68	5.88	0.76
	Feature Subtraction	66.50	50.36	38.02	27.79	23.78	50.07	64.44	15.75	0.38
	Difference Image	65.80	51.21	40.17	31.12	24.01	48.84	68.53	1.27	0.4
CC System based on SVM decoder (PCA)	Feature Concatenation	66.34	51.89	41.34	31.78	25.62	52.12	75.70	5.09	0.75
	Feature Subtraction	68.99	54.18	43.32	33.46	26.47	51.46	72.40	11.51	0.39
	Difference Image	65.62	51.17	39.80	29.98	24.01	50.35	67.86	1.32	0.44

the behavior of the proposed methods according to different combinations of bitemporal images. The achieved results by the two proposed methods are similar to each other. The best combination of bitemporal images is their individual feature subtraction while the worst one is the channelwise difference (DI) of the bitemporal images. This might be attributed to the fact that the backbone is pretrained on ImageNet and as a consequence, it is more natural to have as input unprocessed images rather than the DI. We also noticed that the feature maps of the individual images are more representative compared to the ones of the DI. Indeed, they seem to be complementary to each other. As explained in Section II, the Dubai CCD is composed of multispectral images. In particular, we have used only those spectral bands that are characterized by a spatial resolution of 30 m (visible and infrared channels). Table II reports the results obtained using only the RGB channels and the combination of all the six available bands, to better exploit the spectral information. The latter is achieved through a spectral dimensionality reduction obtained through PCA. In Table II, we can see that both methods have an important gain in terms of accuracy when PCA is applied with respect to the use of only RGB channels. This confirms the importance of spectral information in RS applications. Last two columns of Tables I and II show the training and testing times of the proposed CC methods. In general, one can notice that the CC system based on SVM decoder is much faster to train and to test compared to the CC system based on the RNN. In particular, the testing time to generate the change descriptions for the whole test set is 7–13 times faster on the LEVIR CCD and to 14–25 times faster on the Dubai CCD

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:31:04 UTC from IEEE Xplore. Restrictions apply.

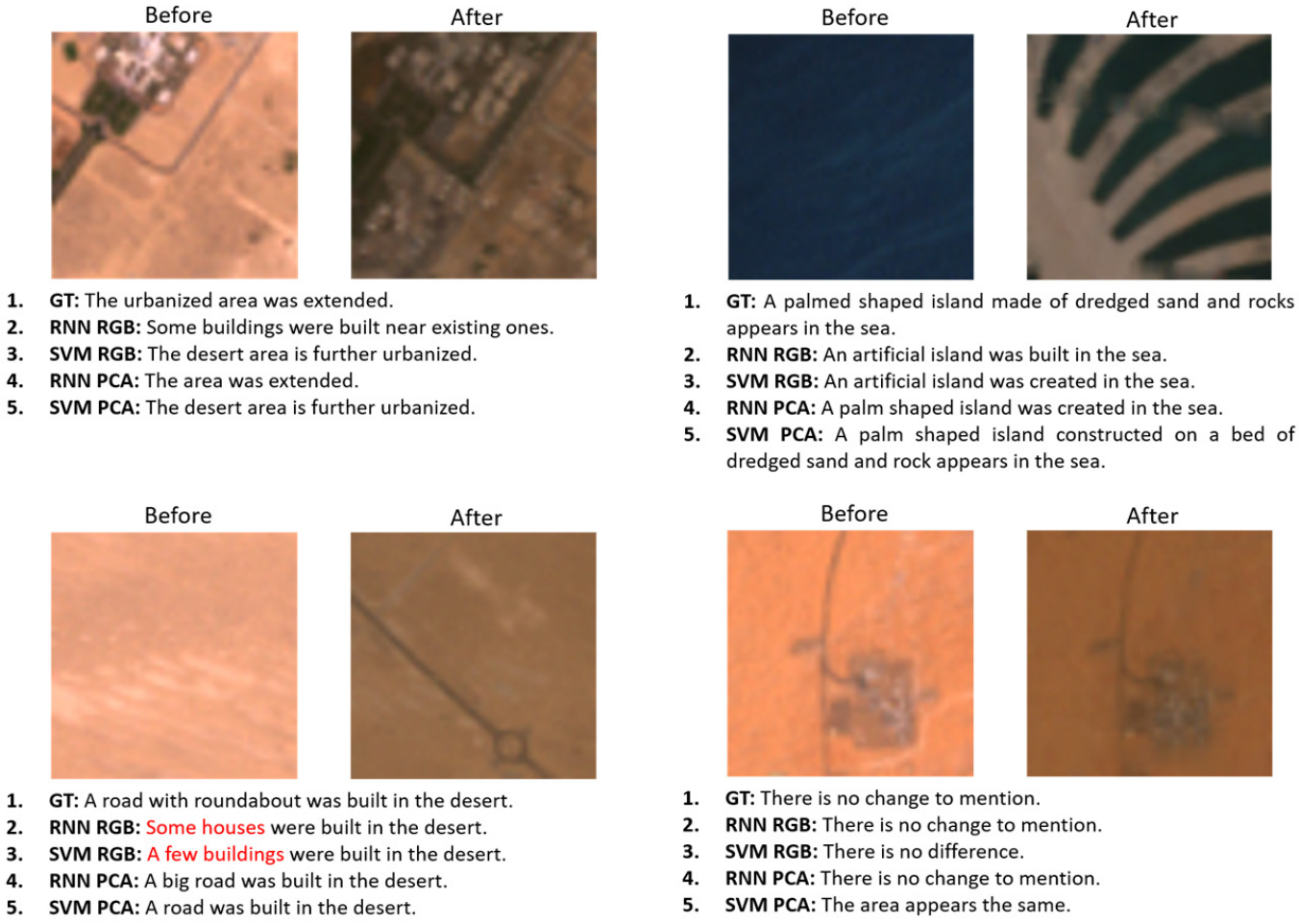


Fig. 7. CC examples of test images from the Dubai CCD. The first description corresponds to one of the ground-truth change descriptions, while the second and third are generated by the CC system based on RNN and SVM, respectively, when subtraction operator is applied to RGB channels. The fourth and fifth descriptions are generated by the CC system based on RNN and SVM, respectively, when the subtraction operator is applied to the three principal components of PCA. Red highlights the errors in the generated change captions.

making it very suitable in real-life scenarios. This is due to the intrinsic buildup of the SVM decoder. In particular, each SVM is responsible to predict a particular word in a certain position up to reaching the recurrent SVM. Hence, the decoder can subdivide the problem into small pieces making it faster compared to the RNN which affront the problem as whole. It is also important to note that the aforementioned quantitative results regarding the CC system based on RNN are calculated on the basis of the model that achieves the highest validation accuracy during training. Hence, the epoch number in which the highest accuracy is reached differs from dataset to dataset as well as from configuration to configuration. In particular, for the RGB Dubai CCD, the epoch number with the highest validation accuracy is 48, 65, and 59 for the feature concatenation, subtraction, and DI fusion operators, respectively; for PCA, Dubai CCD is 40, 53, and 48 feature concatenation, subtraction, and DI fusion operators, respectively; whereas for LEVIR CC is 39, 63, and 75 for feature concatenation, subtraction, and DI fusion operators, respectively.

2) Qualitative Results: Fig. 7 shows four examples of the generated change captions from the proposed systems of bitemporal on Dubai CCD. In particular, we report the qualitative results obtained by the two methods applied to the images considering the RGB channels only and to the three principal components of the PCA that are applied to all the six available bands. The fusion operator considered in these examples is the individual feature subtraction. From the images in Fig. 7, we can notice that all the generated change descriptions generated by the two systems utilizing all the available spectral information are in line with the changes of the bitemporal images. The generated change captions are more complete and coherent with the changes of bitemporal images compared to the ones generated utilizing only the RGB bands. In particular, utilizing only the RGB bands, some generated change descriptions are affected by errors as shown in the first image of the second row in Fig. 7.

Fig. 8 shows two examples of bitemporal images from the LEVIR CCD along with the generated change captions from the two proposed systems. In particular, the generation change captions are shown when the fusion operator is concatenation and subtraction. The generated captions by the two systems are in line with the visual changes between the bitemporal

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:31:04 UTC from IEEE Xplore. Restrictions apply.

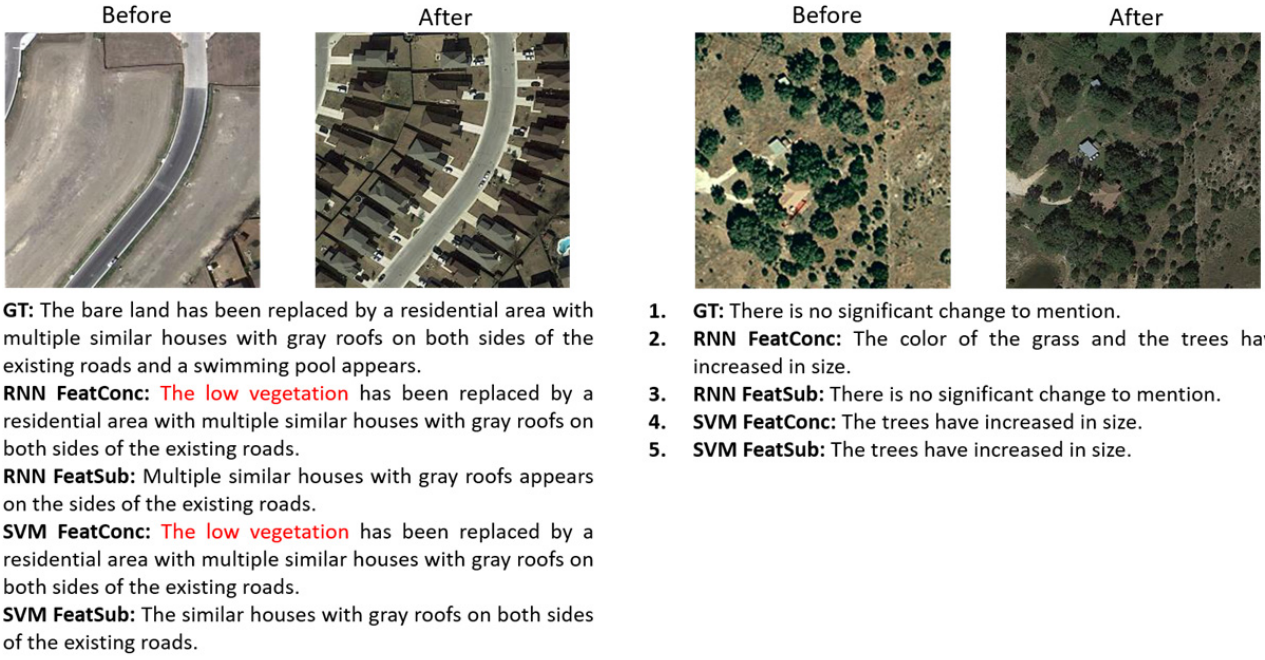


Fig. 8. CC examples of test images from the LEVIR CCD. The first description corresponds to one of the ground-truth change descriptions, while the second and third are generated by the CC system based on RNN when feature concatenation and subtraction operator are applied, respectively. The fourth and fifth descriptions are generated by the CC system based on SVM when feature concatenation and subtraction operator are applied, respectively. Red highlights the errors in the generated change captions.

images. We notice an error in the change generated description of the systems regarding the land-cover transition when concatenation is applied as fusion operator (Fig. 8, first image). In the "before" image, bare land is present instead of the low vegetation. However, the postevent changes are well-described by two methods.

V. Conclusion

In this work, we present two different RSCC systems able at describing the changes from bitemporal images with sentence descriptions. Compared to the traditional CD algorithms, the outputs of the proposed RSCC systems not only are more user-friendly but also include high-level semantic information such as the relationships and the attributes of the detected changes. In the absence of datasets, we constructed two different CCDs. One is based on very high-resolution RGB RS images, and the other one is based on multispectral images. The obtained experimental results show the promising capabilities of the proposed CC systems to generate complete and coherent change sentence descriptions that summarize the changes in bitemporal images. In particular, we show that spectral information is very important in generating reliable change captions. The best strategy to fuse the image information seems to be late fusion where information is integrated at the feature level. From our results, it comes out that early fusion, which is applied at the image level as channelwise difference, is not the best solution for RSCC. This might be explained by the stronger expressive power of deep high-level representations compared to low-level ones. We also showed that the two decoders obtain very similar results in all the metrics. Furthermore, we show that the decoder based on SVMs is faster in terms of training and testing compared to that based on RNN. In the future, we aim at expanding the two proposed datasets with more images. Furthermore, we aim at developing more sophisticated CC methods based on attention mechanisms to better exploit the peculiarities of RS images such as scale and semantic ambiguities as well as spectral information.

References

[1] A. Singh, "Review article digital change detection techniques using remotely-sensed data," *Int. J. Remote Sens.*, vol. 10, no. 6, pp. 989–1003, 1989, doi: 10.1080/01431168908903939.

[2] L. Bruzzone and F. Bovolo, "A novel framework for the design of change-detection systems for very-high-resolution remote sensing images," *Proc. IEEE*, vol. 101, no. 3, pp. 609–630, Mar. 2013, doi: 10.1109/JPROC.2012.2197169.

[3] L. Bruzzone and D. F. Prieto, "Automatic analysis of the difference image for unsupervised change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 3, pp. 1171–1182, May 2000, doi: 10.1109/36.843009.

[4] F. Melgani, G. Moser, and S. B. Serpico, "Unsupervised change-detection methods for remote-sensing images," *Opt. Eng.*, vol. 41, no. 12, pp. 3288–3297, Dec. 2002, doi: 10.1117/1.1518995.

[5] Y. Bazi, L. Bruzzone, and F. Melgani, "An unsupervised approach based on the generalized Gaussian model to automatic change detection in multitemporal SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 874–887, Apr. 2005, doi: 10.1109/TGRS.2004.842441.

[6] F. Melgani and Y. Bazi, "Markovian fusion approach to robust unsupervised change detection in remotely sensed imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 3, no. 4, pp. 457–461, Oct. 2006, doi: 10.1109/LGRS.2006.875773.

[7] G. Moser and S. B. Serpico, "Generalized minimum-error thresholding for unsupervised change detection from SAR amplitude imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 10, pp. 2972–2982, Oct. 2006, doi: 10.1109/TGRS.2006.876288.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:31:04 UTC from IEEE Xplore. Restrictions apply.

[8] T. Celik, "Multiscale change detection in multitemporal satellite images," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 820–824, Oct. 2009, doi: 10.1109/LGRS.2009.2026188.

[9] Y. Bazi, F. Melgani, and H. D. Al-Sharari, "Unsupervised change detection in multispectral remotely sensed imagery with level set methods," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 8, pp. 3178–3187, Aug. 2010, doi: 10.1109/TGRS.2010.2045506.

[10] O. Yousif and Y. Ban, "Improving urban change detection from multitemporal SAR images using PCA-NLM," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 4, pp. 2032–2041, Apr. 2013, doi: 10.1109/TGRS.2013.2245900.

[11] F. Melgani and S. B. Serpico, "A Markov random field approach to spatio-temporal contextual image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 11, pp. 2478–2487, Nov. 2003, doi: 10.1109/TGRS.2003.817269.

[12] P. Serra, X. Pons, and D. Saurí, "Post-classification change detection with data from different sensors: Some accuracy considerations," *Int. J. Remote Sens.*, vol. 24, no. 16, pp. 3311–3340, 2003, doi: 10.1080/0143116021000021189.

[13] F. Yuan, K. E. Sawaya, B. C. Loeffelholz, and M. E. Bauer, "Land cover classification and change analysis of the twin cities (Minnesota) metropolitan area by multitemporal landsat remote sensing," *Remote Sens. Environ.*, vol. 98, nos. 2–3, pp. 317–328, 2005, doi: 10.1016/j.rse.2005.08.006.

[14] P. Zhong and R. Wang, "A multiple conditional random fields ensemble model for urban area detection in remote sensing optical images," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 12, pp. 3978–3988, Dec. 2007, doi: 10.1109/TGRS.2007.907109.

[15] O. Ahlqvist, "Extending post-classification change detection using semantic similarity metrics to overcome class heterogeneity: A study of 1992 and 2001 U.S. National Land Cover Database changes," *Remote Sens. Environ.*, vol. 112, no. 3, pp. 1226–1241, 2008, doi: 10.1016/j.rse.2007.08.012.

[16] C. Benedek and T. Szirányi, "Change detection in optical aerial images by a multilayer conditional mixed Markov model," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 10, pp. 3416–3430, Oct. 2009, doi: 10.1109/TGRS.2009.2022633.

[17] T. Habib, J. Inglada, G. Mercier, and J. Chanussot, "Support vector reduction in SVM algorithm for abrupt change detection in remote sensing," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 3, pp. 606–610, Jul. 2009, doi: 10.1109/LGRS.2009.2020306.

[18] M. Volpi, D. Tuia, F. Bovolo, M. Kanevski, and L. Bruzzone, "Supervised change detection in VHR images using contextual information and support vector machines," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 20, pp. 77–85, Feb. 2013, doi: 10.1016/j.jag.2011.10.013.

[19] P. Singh, Z. Kato, and J. Zerubia, "A multilayer Markovian model for change detection in aerial image pairs with large time differences," in *Proc. 22nd Int. Conf. Pattern Recognit.*, Aug. 2014, pp. 924–929, doi: 10.1109/ICPR.2014.169.

[20] C. Wu, B. Du, X. Cui, and L. Zhang, "A post-classification change detection method based on iterative slow feature analysis and Bayesian soft fusion," *Remote Sens. Environ.*, vol. 199, pp. 241–255, Sep. 2017, doi: 10.1016/j.rse.2017.07.009.

[21] V. Vapnik and V. Vapnik, *Statistical Learning Theory*. New York, NY, USA: Wiley, 1998, pp. 156–160.

[22] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004, doi: 10.1109/TGRS.2004.831865.

[23] Z. Shao, H. Fu, P. Fu, and L. Yin, "Mapping urban impervious surface by fusing optical and SAR data at the decision level," *Remote Sens.*, vol. 8, no. 11, p. 945, Nov. 2016, doi: 10.3390/rs8110945.

[24] N. Ghoggali and F. Melgani, "Genetic SVM approach to semisupervised multitemporal classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 2, pp. 212–216, Apr. 2008, doi: 10.1109/LGRS.2008.915600.

[25] B. Demir, F. Bovolo, and L. Bruzzone, "Updating land-cover maps by classification of image time series: A novel change-detection-driven transfer learning approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 1, pp. 300–312, Jan. 2013, doi: 10.1109/TGRS.2012.2195727.

[26] X. X. Zhu *et al.*, "Deep learning in remote sensing: A comprehensive review and list of resources," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 8–36, Dec. 2017, doi: 10.1109/MGRS.2017.2762307.

[27] L. Ma, Y. Liu, X. Zhang, Y. Ye, G. Yin, and B. A. Johnson, "Deep learning in remote sensing applications: A meta-analysis and review," *ISPRS J. Photogramm. Remote Sens.*, vol. 152, pp. 166–177, Jun. 2019, doi: 10.1016/j.isprsjprs.2019.04.015.

[28] Y. You, J. Cao, and W. Zhou, "A survey of change detection methods based on remote sensing images for multi-source and multi-objective scenarios," *Remote Sens.*, vol. 12, no. 15, p. 2460, Jul. 2020, doi: 10.3390/rs12152460.

[29] L. Khelifi and M. Mignotte, "Deep learning for change detection in remote sensing images: Comprehensive review and meta-analysis," *IEEE Access*, vol. 8, pp. 126385–126400, 2020, doi: 10.1109/ACCESS.2020.3008036.

[30] M. Gong, J. Zhao, J. Liu, Q. Miao, and L. Jiao, "Change detection in synthetic aperture radar images based on deep neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 1, pp. 125–138, Jan. 2016, doi: 10.1109/TNNLS.2015.2435783.

[31] J. Geng, H. Wang, J. Fan, and X. Ma, "Change detection of SAR images based on supervised contractive autoencoders and fuzzy clustering," in *Proc. Int. Workshop Remote Sens. Intell. Process. (RSIP)*, May 2017, pp. 1–3, doi: 10.1109/RSIP.2017.7958819.

[32] C. Zhang, S. Wei, S. Ji, and M. Lu, "Detecting large-scale urban land cover changes from very high resolution remote sensing images using CNN-based classification," *ISPRS Int. J. Geo-Inf.*, vol. 8, no. 4, p. 189, Apr. 2019, doi: 10.3390/ijgi8040189.

[33] S. Saha, F. Bovolo, and L. Bruzzone, "Unsupervised deep change vector analysis for multiple-change detection in VHR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3677–3693, Jun. 2019, doi: 10.1109/TGRS.2018.2886643.

[34] D. Peng, Y. Zhang, and H. Guan, "End-to-end change detection for high resolution satellite images using improved UNet++," *Remote Sens.*, vol. 11, no. 11, p. 1382, Jun. 2019, doi: 10.3390/rs11111382.

[35] M. Yang, L. Jiao, F. Liu, B. Hou, and S. Yang, "Transferred deep learning-based change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6960–6973, Sep. 2019, doi: 10.1109/TGRS.2019.2909781.

[36] R. Hedjam, A. Abdesselam, and F. Melgani, "Change detection in unlabeled optical remote sensing data using Siamese CNN," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4178–4187, 2020, doi: 10.1109/JSTARS.2020.3009116.

[37] Z. Wang, C. Peng, Y. Zhang, N. Wang, and L. Luo, "Fully convolutional Siamese networks based change detection for optical aerial images with focal contrastive loss," *Neurocomputing*, vol. 457, pp. 155–167, Oct. 2021, doi: 10.1016/j.neucom.2021.06.059.

[38] J. Liu, W. Zhang, F. Liu, and L. Xiao, "A probabilistic model based on bipartite convolutional neural network for unsupervised change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, doi: 10.1109/TGRS.2021.3071347.

[39] Z. Li, C. Yan, Y. Sun, and Q. Xin, "A densely attentive refinement network for change detection based on very-high-resolution bitemporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022, doi: 10.1109/TGRS.2022.3159544.

[40] R. Liu, Z. Cheng, L. Zhang, and J. Li, "Remote sensing image change detection based on information transmission and attention mechanism," *IEEE Access*, vol. 7, pp. 156349–156359, 2019, doi: 10.1109/ACCESS.2019.2947286.

[41] B. Qu, X. Li, D. Tao, and X. Lu, "Deep semantic understanding of high resolution remote sensing image," in *Proc. Int. Conf. Comput., Inf. Telecommun. Syst. (CITS)*, Jul. 2016, pp. 1–5, doi: 10.1109/CITS.2016.7546397.

[42] Z. Shi and Z. Zou, "Can a machine generate humanlike language descriptions for a remote sensing image?" *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3623–3634, Jun. 2017, doi: 10.1109/TGRS.2017.2677464.

[43] X. Lu, B. Wang, X. Zheng, and X. Li, "Exploring models and data for remote sensing image caption generation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2183–2195, Apr. 2018, doi: 10.1109/TGRS.2017.2776321.

[44] X. Zhang, X. Wang, X. Tang, H. Zhou, and C. Li, "Description generation for remote sensing images using attribute attention mechanism," *Remote Sens.*, vol. 11, no. 6, p. 612, 2019, doi: 10.3390/rs11060612.

[45] B. Wang, X. Lu, X. Zheng, and X. Li, "Semantic descriptions of high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 8, pp. 1274–1278, Aug. 2019, doi: 10.1109/LGRS.2019.2893772.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:31:04 UTC from IEEE Xplore. Restrictions apply.

[46] W. Huang, Q. Wang, and X. Li, "Denoising-based multiscale feature fusion for remote sensing image captioning," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 3, pp. 436–440, Mar. 2021, doi: 10.1109/LGRS.2020.2980933.

[47] Z. Yuan, X. Li, and Q. Wang, "Exploring multi-level attention and semantic relationship for remote sensing image captioning," *IEEE Access*, vol. 8, pp. 2608–2620, 2020, doi: 10.1109/ACCESS.2019.2962195.

[48] G. Hoxha, F. Melgani, and J. Slaghenauffi, "A new CNN-RNN framework for remote sensing image captioning," in *Proc. Medit. Middle-East Geosci. Remote Sens. Symp. (M2GARSS)*, Mar. 2020, pp. 1–4, doi: 10.1109/M2GARSS47143.2020.9105191.

[49] X. Lu, B. Wang, and X. Zheng, "Sound active attention framework for remote sensing image captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1985–2000, Mar. 2020, doi: 10.1109/TGRS.2019.2951636.

[50] B. Wang, X. Zheng, B. Qu, and X. Lu, "Retrieval topic recurrent memory network for remote sensing image captioning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 256–270, 2020, doi: 10.1109/JSTARS.2019.2959208.

[51] G. Sumbul, S. Nayak, and B. Demir, "SD-RSIC: Summarization-driven deep remote sensing image captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6922–6934, Aug. 2021, doi: 10.1109/TGRS.2020.3031111.

[52] X. Li, X. Zhang, W. Huang, and Q. Wang, "Truncation cross entropy loss for remote sensing image captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5246–5257, Jun. 2021, doi: 10.1109/TGRS.2020.3010106.

[53] G. Hoxha and F. Melgani, "A novel SVM-based decoder for remote sensing image captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, doi: 10.1109/TGRS.2021.3105004.

[54] J. Mao, W. Xu, Y. Yang, J. Wang, and A. L. Yuille, "Explain images with multimodal recurrent neural networks," 2014, *arXiv:1410.1090*.

[55] R. Bernardi *et al.*, "Automatic description generation from images: A survey of models, datasets, and evaluation measures," *J. Artif. Intell. Res.*, vol. 55, pp. 409–442, Jan. 2016.

[56] G. Hoxha, F. Melgani, and B. Demir, "Toward remote sensing image retrieval under a deep image captioning perspective," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4462–4475, 2020, doi: 10.1109/JSTARS.2020.3013818.

[57] M. B. Bejiga, G. Hoxha, and F. Melgani, "Improving text encoding for retro-remote sensing," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 4, pp. 622–626, Apr. 2021, doi: 10.1109/LGRS.2020.2983851.

[58] I. Sutskever, O. Vinyals, and Q. V. Le, "Sequence to sequence learning with neural networks," in *Advances in Neural Information Processing Systems*, Z. Ghahramani, M. Welling, C. Cortes, N. D. Lawrence, and K. Q. Weinberger, Eds. Red Hook, NY, USA: Curran Associates, 2014, pp. 3104–3112. Accessed: Oct. 29, 2019. [Online]. Available: http://papers.nips.cc/paper/5346-sequence-to-sequence-learning-with-neural-networks.pdf

[59] D. Bahdanau, K. Cho, and Y. Bengio, "Neural machine translation by jointly learning to align and translate," 2014, *arXiv:1409.0473*.

[60] J. Sivic and A. Zisserman, "Video Google: A text retrieval approach to object matching in videos," in *Proc. 9th IEEE Int. Conf. Comput. Vis.*, vol. 2, Oct. 2003, pp. 1470–1477, doi: 10.1109/ICCV.2003.1238663.

[61] F. Perronnin, Y. Liu, J. Sánchez, and H. Poirier, "Large-scale image retrieval with compressed Fisher vectors," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2010, pp. 3384–3391.

[62] H. Jégou, F. Perronnin, M. Douze, J. Sánchez, P. Pérez, and C. Schmid, "Aggregating local image descriptors into compact codes," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 9, pp. 1704–1716, Sep. 2011.

[63] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[64] A. See, P. J. Liu, and C. D. Manning, "Get to the point: Summarization with pointer-generator networks," in *Proc. 55th Annu. Meeting Assoc. Comput. Linguistics*, vol. 1, Jul. 2017, pp. 1073–1083, doi: 10.18653/v1/P17-1099.

[65] S. Chouaf, G. Hoxha, Y. Smara, and F. Melgani, "Captioning changes in bi-temporal remote sensing images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 2891–2894, doi: 10.1109/IGARSS47720.2021.9554419.

[66] H. Chen and Z. Shi, "A spatial-temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020, doi: 10.3390/rs12101662.

[67] *Collection-1 Landsat 7 Enhanced Thematic Mapper Plus (ETM+) Level-1 Data Products*, U.S. Geological Survey, Earth Resources Observation and Science (EROS) Center, Reston, VA, USA, 2018, doi: 10.5066/F7WH2P8G.

[68] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255, doi: 10.1109/CVPR.2009.5206848.

[69] M. Tanti, A. Gatt, and K. P. Camilleri, "Where to put the image in an image caption generator," *Natural Lang. Eng.*, vol. 24, no. 3, pp. 467–489, May 2018, doi: 10.1017/S1351324918000098.

[70] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997, doi: 10.1162/neco.1997.9.8.1735.

[71] K. Cho *et al.*, "Learning phrase representations using RNN encoder–decoder for statistical machine translation," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, 2014, pp. 1724–1734, doi: 10.3115/v1/D14-1179.

[72] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," Jan. 2017, *arXiv:1412.6980*. Accessed: Aug. 7, 2022. [Online]. Available: http://arxiv.org/abs/1412.6980

[73] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, "BLEU: A method for automatic evaluation of machine translation," in *Proc. 40th Annu. Meeting Assoc. Comput. Linguistics (ACL)*, 2001, pp. 311–318, doi: 10.3115/1073083.1073135.

[74] C.-Y. Lin, "ROUGE: A package for automatic evaluation of summaries," in *Proc. Workshop Text Summarization Branches Out, Post-Conf. Workshop ACL*, Barcelona, Spain, Jul. 2004, pp. 74–81. [Online]. Available: https://www.microsoft.com/en-us/research/publication/rouge-a-package-for-automatic-evaluation-of-summaries/

[75] S. Banerjee and A. Lavie, "METEOR: An automatic metric for MT evaluation with improved correlation with human judgments," in *Proc. ACL Workshop Intrinsic Extrinsic Eval. Measures Mach. Transl. Summarization*. Ann Arbor, MI, USA, Jun. 2005, pp. 65–72. Accessed: Jul. 20, 2020. [Online]. Available: https://www.aclweb.org/anthology/W05-0909

[76] R. Vedantam, C. L. Zitnick, and D. Parikh, "CIDEr: Consensus-based image description evaluation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 4566–4575, doi: 10.1109/CVPR.2015.7299087.



Genc Hoxha (Graduate Student Member, IEEE) received the B.S. degree in electronics and telecommunications engineering and the M.Sc. degree in telecommunications engineering from the University of Trento, Trento, Italy, in 2014 and 2018, respectively, where he is currently pursuing the Ph.D. degree in signal processing and pattern recognition with the ICT Doctoral School.

His research interests include machine learning and image processing with applications to remote sensing image analysis.



Seloua Chouaf (Member, IEEE) received the State Engineer's degree in automatic control and systems and the Magister's degree in image processing and geographic information systems (TI-SIG) from the University of Science and Technology Houari Boumediene (USTHB), Algiers, Algeria, in 2009 and 2012, respectively, and the Ph.D. degree from the Signal Processing and Recognition (SPR) Laboratory, University of Trento, Trento, Italy, in 2020, within the framework of the Doctoral School of Space Technologies and Applications (EDTAS), an internship abroad.

She is currently an Assistant Professor in electrical engineering and industrial computing (GEII) at the National Superior School of Technology (ENST), Algiers, and a Research Team Member at the Image Processing and Radiation Laboratory (LTIR), USTHB. She took part in different works and projects related to her main areas of research interest that include signal and image processing, pattern recognition, hyperspectral analysis, and remote sensing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:31:04 UTC from IEEE Xplore. Restrictions apply.



Farid Melgani (Fellow, IEEE) received the State Engineer degree in electronics from the University of Batna, Batna, Algeria, in 1994, the M.Sc. degree in electrical engineering from the University of Baghdad, Baghdad, Iraq, in 1999, and the Ph.D. degree in electronic and computer engineering from the University of Genoa, Genoa, Italy, in 2003.

He is currently a Full Professor of Telecommunications with the Department of Information Engineering and Computer Science, University of Trento, Trento, Italy, where he teaches pattern recognition, machine learning, and digital transmission, and is also the Head of the Signal Processing and Recognition Laboratory, the Coordinator of the Doctoral School in Industrial Innovation, and the Dean of Undergrad and Grad Studies. He has coauthored more than 250 scientific publications. His research interests are in the areas of remote sensing, signal/image processing, pattern recognition, machine learning, and computer vision.

Dr. Melgani is currently an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, *International Journal of Remote Sensing*, and IEEE Journal on Miniaturization for Air and Space Systems.



Youcef Smara received the Engineer's degree in electronics from the National Polytechnic School of Algiers, Algiers, Algeria, in 1976, the Magister's degree in image processing from the Institute of Electronics, University of Sciences and Technology Houari Boumediene, Algiers, in 1985, and the Ph.D. degree in image processing and remote sensing from the University of Sciences and Technology Houari Boumediene, in 1998.

He has been a Full Professor and the Director of Research with the University of Sciences and Technology Houari Boumediene, since 2003. He is responsible for the image processing and geographic information system (GIS) option of the Doctoral School of Space Technologies and Applications (EDTAS) in partnership with the Algerian Space Agency (ASAL), Algiers. He has been an expert with the National Geographic Information Council (CNIG), Algiers, since 1998 and he was the Director of the Image Processing and Radiation Laboratory from 2009 to 2013. The main research themes, associated with remote sensing image processing and more specifically radar imagery, data fusion, and hyperspectral imaging, address fundamental problems as well as associated applications.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:31:04 UTC from IEEE Xplore. Restrictions apply.